SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I P. Gowtham Reddy (192525139) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Gowtham Reddy

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

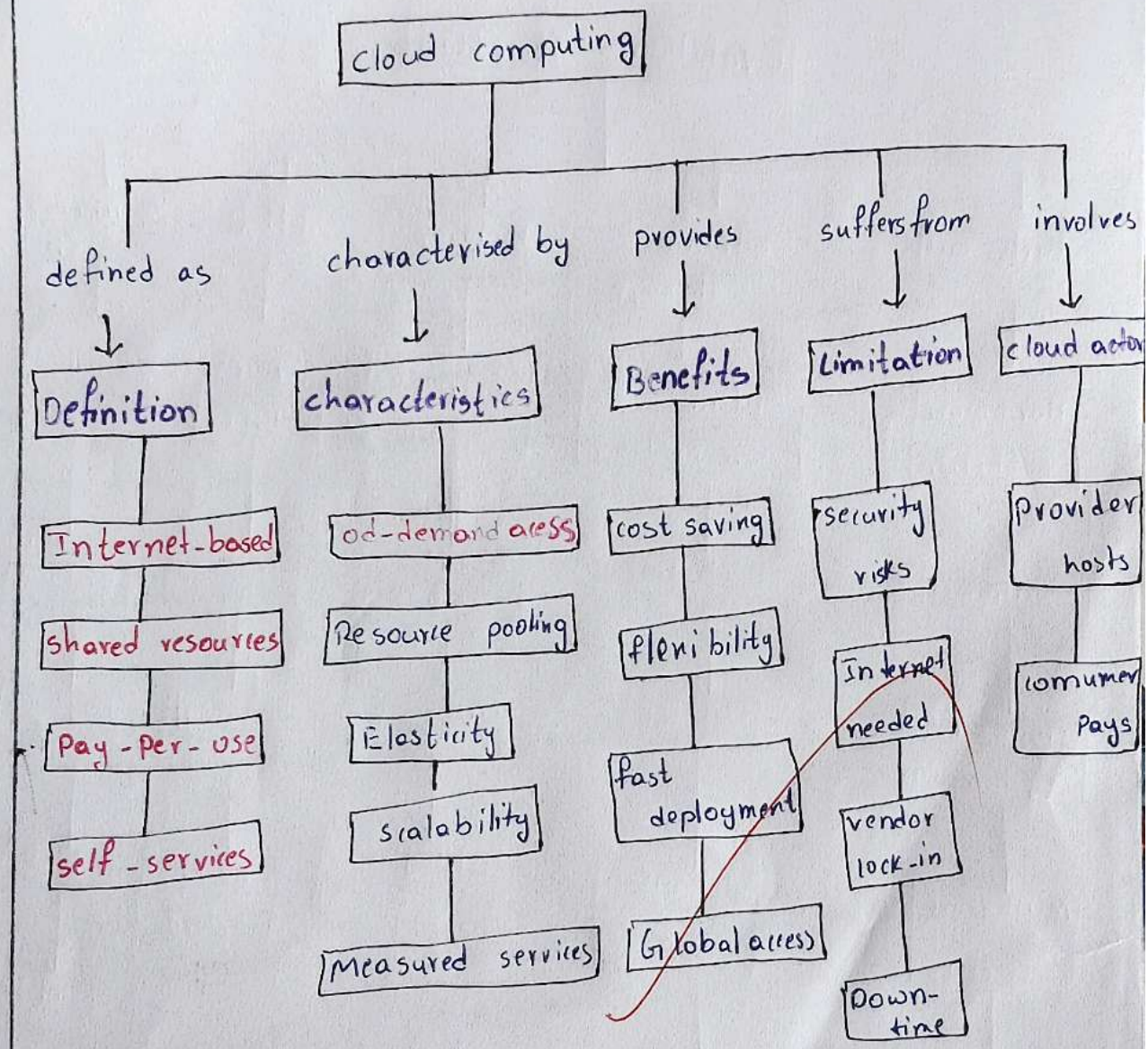
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

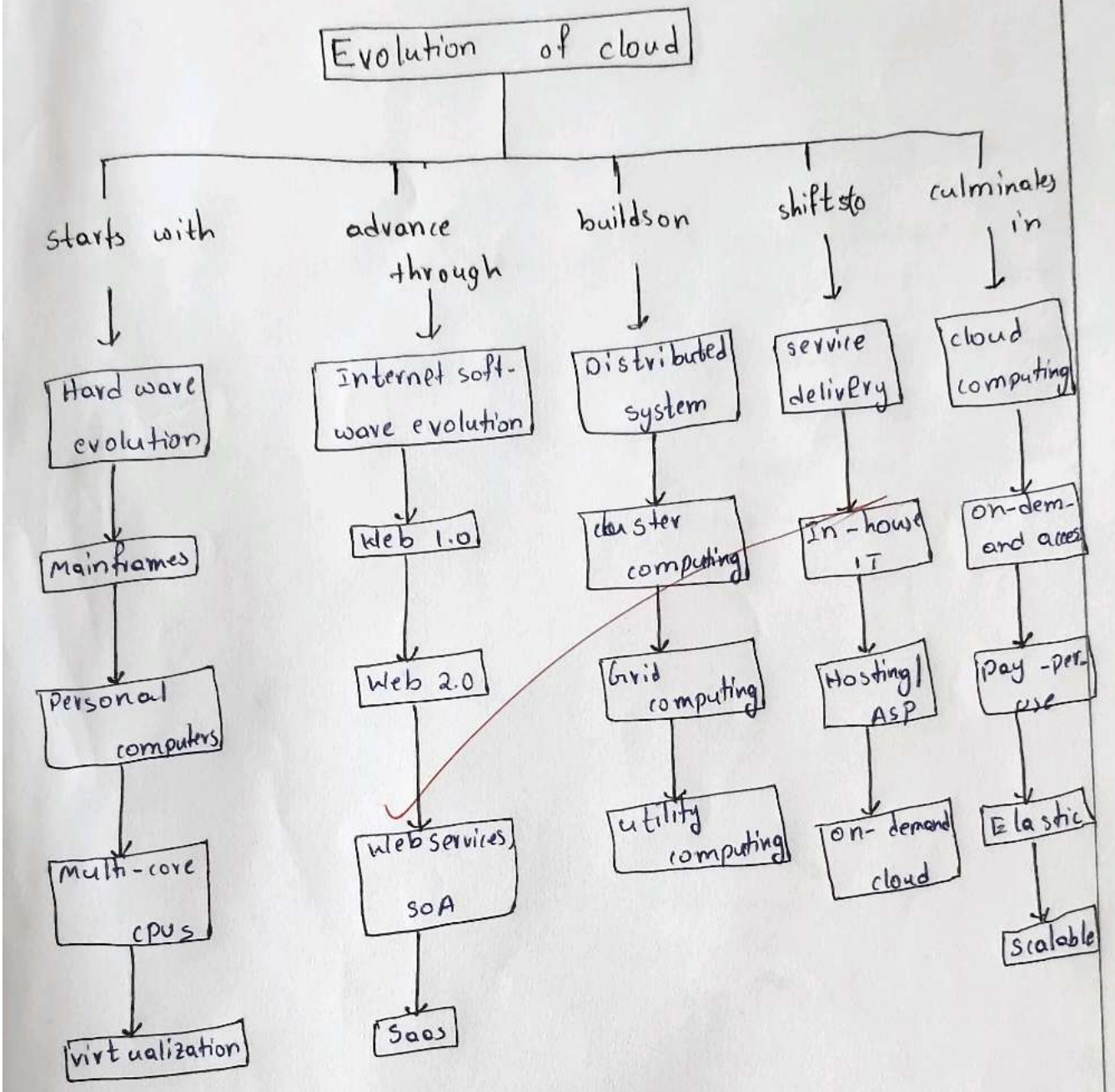
Concept Mapping Rubric

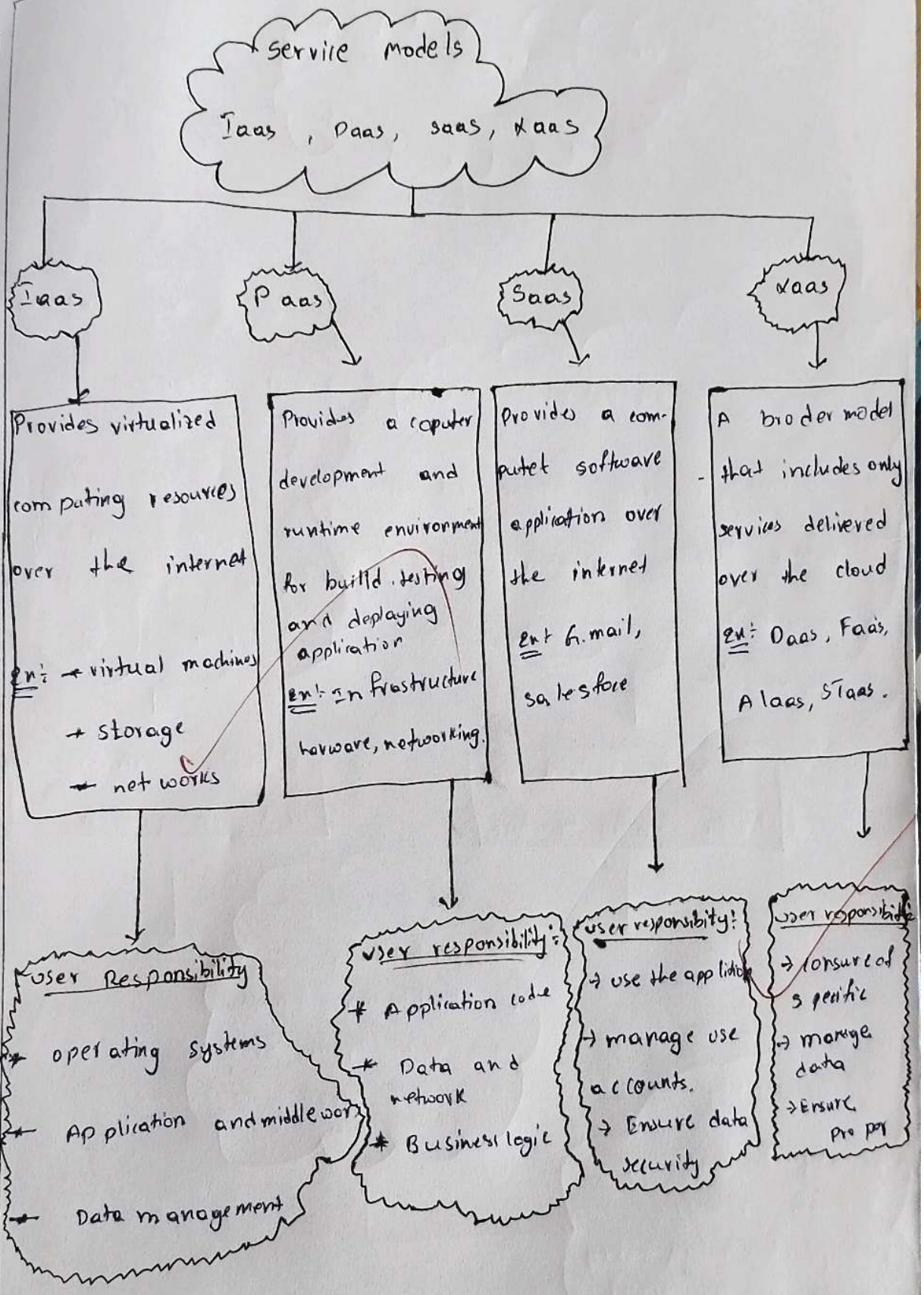
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks:


cloud computing
defined as
characterised by
provides
suffers from
involves
Definition
characteristics
Benefits
Limitation
cloud actor
Internet-based
shared resources
Pay-per-use
self-services
on-demand access
Resource pooling
Elasticity
scalability
Measured services
cost saving
flexibility
fast deployment
Global access
security risks
Internet needed
vendor lock-in
Down-time
Provider hosts
comumer Pays

(2)


Evolution of cloud
starts with
advance through
buildson
shiftsto
culminates in
Hard ware evolution
Internet soft-wave evolution
Distributed system
service delivery
cloud computing
Mainframes
Web 1.0
cluster computing
In-house IT
on-demand access
Personal computers
Web 2.0
Grid computing
Hosting/ ASP
Pay-per-use
Multi-core CPUs
Web services SOA
utility computing
On-demand cloud
Elastic
Virtualization
Saas
Scalable


Servile models
Iaas , Paas, saas, Xaas
Iaas
Paas
Saas
Xaas
Provides virtualized computing resources over the internet
ex:- virtual machines
storage
networks
Provides a coputer development and runtime environment for build. testing and deploying application
ex:- Infrastructure hardware, networking.
Provides a complete software application over the internet
ex:- G.mail, salesfore
A broder model that includes only services delivered over the cloud
ex:- Daas, Faas, Alaas, STaas.
User Responsibility
operating systems
Application and middleware
Data management
User responsibility:
Application code
Data and network
Business logic
user responsibity:
use the applic
manage use accounts.
Ensure data security
user responsibi
ensure of s peciffic
manage data
Ensure Pro pay

(4)

Virtualization and web services as enablers of cloud

Virtulization :

→ creates multiples virtual machines (VMs)

→ Improves resource citization and Isolation

→ Enables flexible allcation of computing resources

web services :

→ users standard protocol [CHTTP, REST, SOAP]

→ Allows different application and system to communicate.

→ Enable easy integration and Interperbility.

Provides isolated and flexible infrostructure

Enable communication and integration

cloud Deployment :

→ virtulization provides the flexible on-demand.

→ web services enable apps and servies to work together.

Scalability

→ easy to add or remove resource

→ supports rapid scaling and elastic growth

13/10/26